

Coding Assignment 3

Due Date: *Monday, October 12 (11.59PM)*

Submission: on Gradescope

Instructions

The coding assignments may be completed in groups of *up to 4 students*. If you do so, please make sure that you include everyone's full name, and that you also *select everyone's name when submitting the assignment on Gradescope*. This ensures that each group member will get a grade assigned and have access to the comments from the graders.

For this assignment, the following items need to be submitted:

1. A **single PDF** that contains your code, all results, plots, and written answers (generated from R Markdown, Jupyter, etc.).
2. The **source file** used to produce the PDF (.Rmd, ipynb, ...).

This coding assignment uses the `CollegeScores4yr` table in the R package `Lock5Data` (U.S. Department of Education College Scorecard). Each row is a U.S. institution that primarily grants bachelor's degrees. The **response** is `CompRate`: *the share of students who finish a program within 150% of normal time*.

Predictors are institutional attributes such as public / private / for-profit `Control`, `Locale`, `Region`, admission rate, test scores, enrollment, student demographics, net price, faculty salary, instructional spending, and Pell share.

The *main task* is to use **tree-based methods** to predict completion (`CompRate`), compare them to simple baselines and one linear model, and then write a short discussion on eight test-set schools.

Load the data as follows.

R:

```
install.packages("Lock5Data")
library(Lock5Data)
data(CollegeScores4yr)
```

Python: Download the `CollegeScores4yr.csv` file from Coursera and load it in Python.

The data is quite messy, so please take the following steps before splitting into training/testing sets:

- Drop rows with missing `CompRate`.
- Drop `Accred` (too many agencies) and `State` (use `Region` for geography).
- Drop `MainDegree`. It is 3 for every school in `CollegeScores4yr`, so it is not a predictor. Leaving it in will cause `lm` to fail.
- Keep `Name` and ID attached to the same rows when you split, but *do not put them in the model*.
- Add missingness indicators for `AdmitRate`, `MidACT`, and `AvgSAT` (1 if that field is missing, 0 otherwise). Test scores and admit rates are not missing at random.
- Latitude and longitude are optional; they are not needed for a good model.

After the split (training statistics only):

- For every remaining numeric predictor, fill missing values with the *training* median. Apply those same medians to the test set.
- For every remaining categorical predictor, fill missing values with the *training* mode. If the test set contains a factor level that never appears in training, map it to that mode.
- Do not drop additional rows, and do not compute fill-in values from the test set or from the combined data.

Part I: Tree-based methods

- (a) Split the data into training (70%) and testing (30%) data sets. Use seed 598 for reproducibility of the results. Keep `Name` and `ID` aligned with the same rows.

After the split, and after the training-only fill-in, compute two *baselines* on the test set:

(i) the training-mean of `CompRate` for every test row; (ii) the training-mean of `CompRate` *within* each `Control` level. *Report both MSEs.*

- (b) Fit a tree model and visualize it. Interpret at least two splits. Say whether each split looks more like student mix (Pell, demographics), selectivity (admit rate, SAT/ACT), or resources (faculty salary, instructional spending).

- (c) Prune the tree using cross-validation. Compare test MSE of the pruned vs. unpruned tree. Then grow a *second* tree using only the **public** training rows. Do the first two splits match the pooled tree? If not, the pooled tree is averaging different kinds of institution together. Comment on what changed.
- (d) Fit a random forest model to the training data. Choose two different values for `mtry` (in R) or `max_features` (in Python). If you use R, set `importance=TRUE`. In Python this is automatic with `RandomForestRegressor`. Comment on the results.
- (e) From the forest in (d): which predictors look most important? Also name **one** predictor that is important in the forest but barely appears in the single tree from (b)–(c), and give a one-sentence reason that can happen.
- (f) Report the test MSE for the forest and compare it with the pruned tree. Also report test MSE *by Control* (Public / Private / Profit) for both models. What do you observe?
- (g) Fit a GBM. Choose at least two learning rates (shrinkage). In R use `distribution="gaussian"` in `gbm`. Choose the number of trees by cross-validation on the *training* data. Evaluate that choice once on the test set. You may plot test MSE versus number of trees for the two learning rates as a diagnostic of over/underfitting; do not use that plot to pick the number of trees. *Interpret the two most influential variables.*
- (h) Fit one linear baseline on encoded features (ordinary least squares or ridge is enough). Encode categoricals and standardize numerics using the training set only. **Compare test MSE** of: mean baseline, Control-mean baseline, pruned tree, forest, GBM, and the linear model. Comment.

Part II: Eight schools

From your **test** set only, choose eight schools and print them: **3 Public**, **3 Private**, and **2 Profit**. Do not pick the eight schools by looking at the residuals or by observed completion. Training-set rows are not allowed.

Then create the following table: **Name**, **Control**, **Pell**, predicted **CompRate** from your preferred Part I model, observed **CompRate**, residual (observed minus predicted).

Based on your results and the models from Part I, answer the following questions (one or two sentences is sufficient):

- (1) Which of the eight schools has the largest absolute residual? Did the model over- or under-predict, and what kind of school is it (using **Control** and the name)?
- (2) Do the errors in your table line up with **Control**, with missing test scores, or with neither?
- (3) These eight numbers are predictions, not a ranking of school quality and not an effect of changing Pell, spending, or admissions. In one sentence, *state a conclusion someone might draw from your table that the model does not support.*